

Linear Models

We apply the linear estimation theory of the previous two chapters to the important special case of linear models, which arises often in applications. Specifically, we now assume that the zero-mean random vectors $\{x, y\}$ are related via a linear model of the form

$$y = Hx + v \quad (5.1)$$

for some $q \times p$ matrix H . Here v denotes a zero-mean random noise vector with known covariance matrix, $R_v = E vv^*$. The covariance matrix of x is also assumed to be known, say, $E xx^* = R_x$. Both $\{x, v\}$ are uncorrelated, i.e., $E xv^* = 0$, and we further assume that $R_x > 0$ and $R_v > 0$.

5.1 ESTIMATION USING LINEAR RELATIONS

According to Thm. 3.1, when $R_y > 0$, the linear least-mean-squares estimator of x given y is

$$\hat{x} = R_{xy} R_y^{-1} y \quad (5.2)$$

Because of (5.1), the covariances $\{R_{xy}, R_y\}$ can be determined in terms of the given matrices $\{H, R_x, R_v\}$. Indeed, the uncorrelatedness of $\{x, v\}$ gives

$$\begin{aligned} R_y &= E yy^* = E (Hx + v)(Hx + v)^* = HR_x H^* + R_v \\ R_{xy} &= E xy^* = E x(Hx + v)^* = R_x H^* \end{aligned}$$

Moreover, since $R_v > 0$ we get $R_y > 0$. The expression (5.2) for $\hat{x}$ then becomes

$$\hat{x} = R_x H^* [R_v + H R_x H^*]^{-1} y \quad (5.3)$$

This expression can be rewritten in an equivalent form by using the so-called *matrix inversion formula* or *lemma*. This formula is a very useful matrix theory result and it will be called upon several times throughout this book. The result states that for arbitrary matrices $\{A, B, C, D\}$ of compatible dimensions, if A and C are invertible, then

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1} \quad (5.4)$$

The identity can be verified algebraically; it essentially shows how the inverse of the sum $A + BCD$ is related to the inverse of A .

Applying (5.4) to the matrix $[R_v + H R_x H^*]^{-1}$ in (5.3), with the identifications

$$A = R_v, \quad B = H, \quad C = R_x, \quad D = H^*$$

we obtain

$$\begin{aligned}\hat{x} &= R_x H^* \{R_v^{-1} - R_v^{-1} H (R_x^{-1} + H^* R_v^{-1} H)^{-1} H^* R_v^{-1}\} \mathbf{y} \\ &= \{R_x (R_x^{-1} + H^* R_v^{-1} H) - R_x H^* R_v^{-1} H\} (R_x^{-1} + H^* R_v^{-1} H)^{-1} H^* R_v^{-1} \mathbf{y} \\ &= [R_x^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} \mathbf{y}\end{aligned}$$

where in the second equality we factored out the term $(R_x^{-1} + H^* R_v^{-1} H)^{-1} H^* R_v^{-1} \mathbf{y}$ from the right. Hence,

$$\hat{x} = [R_x^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} \mathbf{y} \quad (5.5)$$

This alternative form can be useful in several contexts. Observe, for example, that when H is a column vector, the quantity $(R_v + H R_x H^*)$ that appears in (5.3) is a matrix, while the quantity $(R_x^{-1} + H^* R_v^{-1} H)$ that appears in (5.5) is a scalar. In this case, the representation (5.5) leads to a simpler expression for $\hat{x}$. In general, the convenience of using (5.3) or (5.5) depends on the situation at hand.

It further follows that the m.m.s.e. matrix is given by

$$\begin{aligned}\text{m.m.s.e.} = E \tilde{x} \tilde{x}^* &= E (x - \hat{x}) x^*, \quad \text{since } \tilde{x} \perp \hat{x} \\ &= R_x - [R_x^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} H R_x \\ &= [R_x^{-1} + H^* R_v^{-1} H]^{-1}\end{aligned}$$

where in the last equality we used the matrix inversion lemma again. That is,

$$\text{m.m.s.e.} = [R_x^{-1} + H^* R_v^{-1} H]^{-1}$$

Theorem 5.1 (Linear estimator for linear models) Let $\{y, x, v\}$ be zero-mean random variables that are related via the linear model $y = Hx + v$, for some data matrix H of compatible dimensions. Both x and v are assumed uncorrelated with invertible covariance matrices, $R_v = E v v^*$ and $R_x = E x x^*$. The linear least-mean-squares estimator of x given y can be evaluated by either expression:

$$\hat{x} = R_x H^* [R_v + H R_x H^*]^{-1} y = [R_x^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} y$$

and the resulting minimum mean-square error matrix is

$$\text{m.m.s.e.} = [R_x^{-1} + H^* R_v^{-1} H]^{-1}$$

Remark 5.1 (Centering for linear models) If the variables $\{x, v\}$ in (5.1) were not zero-mean, say, $E x = \bar{x}$ and $E v = \bar{v}$, then the above results will still hold with $\hat{x}$ replaced by $(\hat{x} - \bar{x})$ and y replaced by $(y - \bar{y}) = (y - H\bar{x} - \bar{v})$. Indeed, the covariance matrices $\{R_x, R_v, R_y\}$ will need to be defined accordingly as

$$\begin{aligned}R_x &= E (x - \bar{x})(x - \bar{x})^*, \quad R_v = E (v - \bar{v})(v - \bar{v})^* \\ R_y &= E (y - \bar{y})(y - \bar{y})^*, \quad R_{xy} = E (x - \bar{x})(y - \bar{y})^*\end{aligned}$$

and the uncorrelatedness of $\{x, v\}$ will now amount to requiring $\mathbf{E}(x - \bar{x})(v - \bar{v})^* = 0$. Under these conditions, it will still hold that $R_{xy} = R_x H^*$ and $R_y = R_v + H R_x H^*$, and the expressions for $\hat{x}$ will become (cf. (4.13)):

$$\begin{aligned}\hat{x} &= \bar{x} + [R_x^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} (y - H\bar{x} - \bar{v}) \\ &= \bar{x} + R_x H^* [H R_x H^* + R_v]^{-1} (y - H\bar{x} - \bar{v})\end{aligned}$$

◇

We now illustrate the application of Thm. 5.1 to several important examples including channel estimation, channel equalization, and block data estimation.

5.2 APPLICATION: CHANNEL ESTIMATION

Consider an FIR channel whose tap vector c is unknown; it is modeled as a zero-mean random vector with a known covariance matrix, $R_c = \mathbf{E} c c^*$. The following experiment is performed with the purpose of estimating c , assumed of length M . The channel is assumed initially at rest (i.e., no initial conditions in its delay elements) and a known input sequence $\{s(i)\}$, also called a *training* sequence, is applied to the channel. The resulting output sequence $\{z(i)\}$ is measured in the presence of additive noise, $v(i)$, as shown in Fig. 5.1. The available measurements are

$$y(i) = z(i) + v(i) \quad (5.6)$$

where $v(i)$ is a zero-mean noise sequence that is uncorrelated with c .

Assume we collect a block of measurements $\{s(\cdot), y(\cdot)\}$ over the interval $0 \leq i \leq N$. Then we can write in matrix form, say, for $M = 3$ and $N = 6$,

$$\underbrace{\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \\ y(5) \\ y(6) \end{bmatrix}}_{y: (N+1) \times 1} = \underbrace{\begin{bmatrix} s(0) & & \\ s(1) & s(0) & \\ s(2) & s(1) & s(0) \\ s(3) & s(2) & s(1) \\ s(4) & s(3) & s(2) \\ s(5) & s(4) & s(3) \\ s(6) & s(5) & s(4) \end{bmatrix}}_{H: (N+1) \times M} c + \underbrace{\begin{bmatrix} v(0) \\ v(1) \\ v(2) \\ v(3) \\ v(4) \\ v(5) \\ v(6) \end{bmatrix}}_{v: (N+1) \times 1} \quad (5.7)$$

where we are further defining the quantities $\{y, H, v\}$. Note that the *data matrix* H has a rectangular Toeplitz structure, i.e., it has constant entries along its diagonals. In addition, each row of H amounts to a state vector (also called a regressor) of the FIR channel.

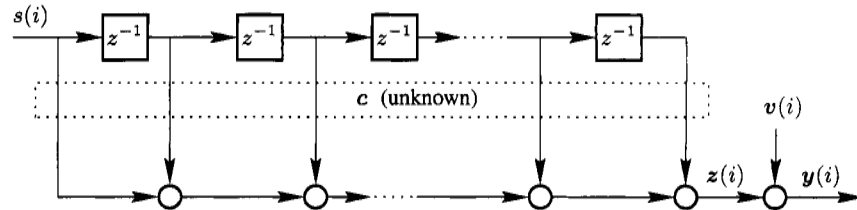


FIGURE 5.1 Channel estimation in the presence of additive noise.

Specifically, the i -th row of H has the form

$$[s(i) \quad s(i-1) \quad \dots \quad s(i-M+1)]$$

which contains the input at time i , $s(i)$, as well as the outputs of all delay elements in the channel. Thus, it is common to refer to the i -th row of H as the state vector or the regressor of the channel at the time instant i .

The quantities $\{\mathbf{y}, H\}$ so defined are both available to the designer, in addition to the covariance matrices $\{R_c = E c c^*, R_v = E v v^*\}$ (by assumption). In particular, if the noise sequence $\{v(i)\}$ is assumed white with variance σ_v^2 , then $R_v = E v v^* = \sigma_v^2 I$. With this information, we can estimate the channel as follows. Since we have a linear model relating $\mathbf{y}$ to c , as indicated by (5.7), then according to Thm. 5.1, the optimal linear estimator for c can be obtained from either expression:

$$\hat{c} = R_c H^* [H R_c H^* + R_v]^{-1} \mathbf{y} = [R_c^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} \mathbf{y} \quad (5.8)$$

5.3 APPLICATION: BLOCK DATA ESTIMATION

Our second application is in the context of data (or symbol) recovery. We consider the same FIR channel as in Fig. 5.1, except that now we assume that its tap vector is known. For example, it could have been estimated via a prior training procedure as explained in the previous section. We denote this tap vector by c , with individual entries

$$c \triangleq \text{col}\{c(0), c(1), \dots, c(M-1)\}$$

The channel is initially at rest and its output sequence, $\{z(i)\}$, is again measured in the presence of additive noise, $v(i)$, as shown in Fig. 5.2. The signals $\{v(\cdot), s(\cdot)\}$ are assumed uncorrelated. What we would like to estimate now are the symbols $\{s(\cdot)\}$ that are being transmitted through the channel. Observe that, to be consistent with our notation, since c is deterministic and $s(\cdot)$ is random, we are now using a boldface letter for the latter and a normal font for the former.

Suppose we collect a block of measurements $\{\mathbf{y}(\cdot)\}$, say, $(N+1)$ measurements over the interval $0 \leq i \leq N$. Rather than relate the $\{\mathbf{y}(\cdot)\}$ to the channel tap vector c through a data matrix, as we did in (5.7), we now relate the $\{\mathbf{y}(\cdot)\}$ to the $\{s(\cdot)\}$ through a *channel matrix*. More specifically, assume again that $M = 3$ and $N = 6$ for illustration purposes.

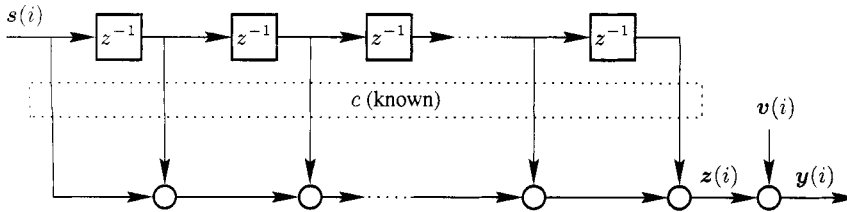


FIGURE 5.2 Block data estimation in the presence of additive noise.

Then we can write

$$\underbrace{\begin{bmatrix} \mathbf{y}(0) \\ \mathbf{y}(1) \\ \mathbf{y}(2) \\ \mathbf{y}(3) \\ \mathbf{y}(4) \\ \mathbf{y}(5) \\ \mathbf{y}(6) \end{bmatrix}}_{\mathbf{y}: (N+1) \times 1} = \underbrace{\begin{bmatrix} c(0) & & & & & & \\ c(1) & c(0) & & & & & \\ c(2) & c(1) & c(0) & & & & \\ & c(2) & c(1) & c(0) & & & \\ & & c(2) & c(1) & c(0) & & \\ & & & c(2) & c(1) & c(0) & \\ & & & & c(2) & c(1) & c(0) \end{bmatrix}}_{H: (N+1) \times (N+1)} \underbrace{\begin{bmatrix} \mathbf{s}(0) \\ \mathbf{s}(1) \\ \mathbf{s}(2) \\ \mathbf{s}(3) \\ \mathbf{s}(4) \\ \mathbf{s}(5) \\ \mathbf{s}(6) \end{bmatrix}}_{\mathbf{s}: (N+1) \times 1} + \underbrace{\begin{bmatrix} \mathbf{v}(0) \\ \mathbf{v}(1) \\ \mathbf{v}(2) \\ \mathbf{v}(3) \\ \mathbf{v}(4) \\ \mathbf{v}(5) \\ \mathbf{v}(6) \end{bmatrix}}_{\mathbf{v}: (N+1) \times 1}$$

Note that the channel matrix H is now square Toeplitz of size $(N+1) \times (N+1)$; it also has a banded structure. The quantities $\{\mathbf{y}, H\}$ so defined are available to the designer, in addition to the covariance matrices $\{R_s, R_v\}$ (by assumption). In particular, if the data and noise sequences $\{\mathbf{s}(\cdot), \mathbf{v}(\cdot)\}$ are white with variances $\{\sigma_s^2, \sigma_v^2\}$, then $R_s = E \mathbf{s} \mathbf{s}^* = \sigma_s^2 \mathbf{I}$ and $R_v = E \mathbf{v} \mathbf{v}^* = \sigma_v^2 \mathbf{I}$. With this information, we can estimate the symbols in the vector $\mathbf{s}$ as follows. Observe again that we have a linear model relating $\mathbf{y}$ to the unknown symbol vector $\mathbf{s}$. According to Thm. 5.1, the optimal linear estimator for $\mathbf{s}$ can then be found from either expression:

$$\hat{\mathbf{s}} = R_s H^* [H R_s H^* + R_v]^{-1} \mathbf{y} = [R_s^{-1} + H^* R_v^{-1} H]^{-1} H^* R_v^{-1} \mathbf{y} \quad (5.9)$$

5.4 APPLICATION: LINEAR CHANNEL EQUALIZATION

Our third application is in the context of linear channel equalization. More specifically, we generalize the discussion of Ex. 4.1.

Consider again an FIR channel as shown in Fig. 5.2, with a known tap vector c of length M . Data symbols $\{\mathbf{s}(\cdot)\}$ are transmitted through the channel and the output sequence, $\{\mathbf{z}(i)\}$, is measured in the presence of additive noise, $\mathbf{v}(i)$. The signals $\{\mathbf{v}(\cdot), \mathbf{s}(\cdot)\}$ are assumed uncorrelated. Due to channel memory, each measurement $\mathbf{y}(i)$ contains contributions not only from $\mathbf{s}(i)$ but also from prior symbols since

$$\mathbf{y}(i) = c(0)\mathbf{s}(i) + \underbrace{\sum_{k=1}^{M-1} c(k)\mathbf{s}(i-k)}_{\text{ISI}} + \mathbf{v}(i) \quad (5.10)$$

The second term on the right-hand side is termed *inter-symbol-interference* (ISI); it refers to the interference that is caused by prior symbols in $\mathbf{y}(i)$. The purpose of an equalizer is to recover $\mathbf{s}(i)$. To achieve this task, an equalizer does not only rely on the most recent measurement $\mathbf{y}(i)$, but it also employs several prior measurements $\{\mathbf{y}(i-k)\}$, say, for $k = 1, 2, \dots, L-1$. These prior measurements contain information that is correlated with the ISI term in $\mathbf{y}(i)$. For example, the expression for $\mathbf{y}(i-1)$ is

$$\mathbf{y}(i-1) = c(0)\mathbf{s}(i-1) + \underbrace{\sum_{k=1}^{M-1} c(k)\mathbf{s}(i-1-k)}_{\text{ISI}} + \mathbf{v}(i-1)$$

and the ISI term in it shares several data symbols with the ISI term in $\mathbf{y}(i)$. It is for this reason that prior measurements are useful in eliminating (or reducing) ISI. In this section,

we are interested in an equalizer structure of the form shown in Fig. 5.3. The equalizer is chosen to have an FIR structure with L coefficients, so that for each time instant i , it would employ the L observations

$$\mathbf{y}_i \triangleq \begin{bmatrix} \mathbf{y}(i) \\ \mathbf{y}(i-1) \\ \mathbf{y}(i-2) \\ \vdots \\ \mathbf{y}(i-L+1) \end{bmatrix}$$

in order to estimate $s(i-\Delta)$ for some integer delay $\Delta \geq 0$. We remark that we shall frequently deal with *time sequences* in this book. So when we write $\mathbf{y}(i)$ we are referring to the value of the time sequence $\mathbf{y}(\cdot)$ at time i . Not only that, but the notation $\mathbf{y}(\cdot)$, with parentheses, also means that $\mathbf{y}(\cdot)$ is a scalar. This is because for vector-valued time-sequences, we shall instead write $\mathbf{y}_i$, with a subscript rather than parentheses, to refer to its value at time i . In other words, we shall use parenthesis for *time-indexing* in the scalar case, e.g., $\{\mathbf{y}(i)\}$, and subscripts in the vector case, e.g., $\{\mathbf{y}_i\}$.

It is useful to express the observation vector $\mathbf{y}_i$ in terms of the transmitted data as follows. Assume for the sake of illustration that $L = 5$ (i.e., an equalizer with 5 taps) and $M = 4$ (a channel with four taps). Then we have

$$\underbrace{\begin{bmatrix} \mathbf{y}(i) \\ \mathbf{y}(i-1) \\ \mathbf{y}(i-2) \\ \mathbf{y}(i-3) \\ \mathbf{y}(i-4) \end{bmatrix}}_{\mathbf{y}_i: L \times 1} = \underbrace{\begin{bmatrix} c(0) & c(1) & c(2) & c(3) & & & \\ & c(0) & c(1) & c(2) & c(3) & & \\ & & c(0) & c(1) & c(2) & c(3) & \\ & & & c(0) & c(1) & c(2) & c(3) \\ & & & & c(0) & c(1) & c(2) & c(3) \end{bmatrix}}_{H: L \times (L+M-1)} \underbrace{\begin{bmatrix} s(i) \\ s(i-1) \\ s(i-2) \\ s(i-3) \\ s(i-4) \\ s(i-5) \\ s(i-6) \\ s(i-7) \end{bmatrix}}_{\mathbf{s}_i: (L+M)-1 \times 1} + \underbrace{\begin{bmatrix} \mathbf{v}(i) \\ \mathbf{v}(i-1) \\ \mathbf{v}(i-2) \\ \mathbf{v}(i-3) \\ \mathbf{v}(i-4) \end{bmatrix}}_{\mathbf{v}_i: L \times 1}$$

Note that the observation vector $\mathbf{y}_i$ has L entries, the data vector $\mathbf{s}_i$ has $L+M-1$ entries, and the channel matrix is now $L \times (L+M-1)$. We thus find that there is a linear relation between the vectors $\{\mathbf{y}_i, \mathbf{s}_i\}$ and this relation can be used to evaluate the covariance and cross-covariance quantities that are needed to estimate $s(i-\Delta)$ from $\mathbf{y}_i$ in the linear least-mean-squares sense. Specifically, let us write

$$\hat{s}(i-\Delta) = \mathbf{w}^* \mathbf{y}_i \quad (5.11)$$

for some column vector $\mathbf{w}$ to be determined. We denote the optimal choice for $\mathbf{w}$ by $\mathbf{w}^o$; the entries of $\mathbf{w}^{o*}$ will correspond to the optimal tap coefficients for the equalizer. According to Thm. 3.1, $\mathbf{w}^{o*}$ is given by

$$\mathbf{w}^{o*} = \mathbf{R}_{s\mathbf{y}} \mathbf{R}_{\mathbf{y}}^{-1} \quad (5.12)$$

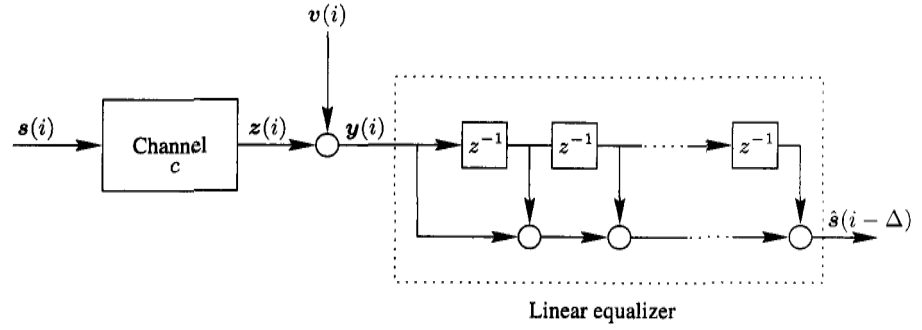


FIGURE 5.3 Linear equalization of an FIR channel in the presence of additive noise.

where

$$R_{sy} \triangleq \mathbf{E} \mathbf{s}(i - \Delta) \mathbf{y}_i^* \quad (1 \times L) \quad (5.13)$$

denotes the cross-covariance vector between $\mathbf{s}(i - \Delta)$ and $\mathbf{y}_i$, and

$$R_y \triangleq \mathbf{E} \mathbf{y}_i \mathbf{y}_i^* \quad (L \times L) \quad (5.14)$$

denotes the covariance matrix of the observation vector, $\mathbf{y}_i$. Observe that since the processes $\{\mathbf{s}(\cdot), \mathbf{y}(\cdot)\}$ are jointly wide-sense stationary, the quantities $\{R_{sy}, R_y\}$ are independent of i . In order to determine $\{R_{sy}, R_y\}$ we resort to the aforementioned linear model relating $\{\mathbf{y}_i, \mathbf{s}_i, \mathbf{v}_i, H\}$. To begin with,

$$R_y = \mathbf{E} (H \mathbf{s}_i + \mathbf{v}_i) (H \mathbf{s}_i + \mathbf{v}_i)^* = H R_s H^* + R_v$$

where

$$R_s \triangleq \mathbf{E} \mathbf{s}_i \mathbf{s}_i^* \quad ((L + M - 1) \times (L + M - 1)) \quad \text{and} \quad R_v \triangleq \mathbf{E} \mathbf{v}_i \mathbf{v}_i^* \quad (L \times L)$$

and

$$R_{sy} = \mathbf{E} \mathbf{s}(i - \Delta) (H \mathbf{s}_i + \mathbf{v}_i)^* = (\mathbf{E} \mathbf{s}(i - \Delta) \mathbf{s}_i^*) H^*$$

since $\{\mathbf{v}(\cdot), \mathbf{s}(\cdot)\}$ are uncorrelated. The value of $\mathbf{E} \mathbf{s}(i - \Delta) \mathbf{s}_i^*$ depends on the assumed correlation between the transmitted symbols. If the $\{\mathbf{s}(\cdot)\}$ are independent and identically distributed with variance σ_s^2 , then $R_s = \sigma_s^2 \mathbf{I}$ and

$$\begin{aligned} \mathbf{E} \mathbf{s}(i - \Delta) \mathbf{s}_i^* &= \mathbf{E} \mathbf{s}(i - \Delta) \begin{bmatrix} \mathbf{s}^*(i) & \mathbf{s}^*(i - 1) & \dots & \mathbf{s}^*(i - 7) \end{bmatrix} \\ &= \begin{bmatrix} 0 & \dots & 0 & \sigma_s^2 & 0 & \dots & 0 \end{bmatrix} \end{aligned}$$

with Δ leading zeros. We continue with this assumption on the data $\{\mathbf{s}(\cdot)\}$ for simplicity of presentation. But it should be clear that the above derivation applies even for correlated (but stationary) data. In a similar vein, we assume that the noise sequence $\{\mathbf{v}(\cdot)\}$ is white with variance σ_v^2 so that $R_v = \sigma_v^2 \mathbf{I}$. Again, the development applies even for correlated (but stationary) noise. We therefore find that

$$R_y = \sigma_s^2 H H^* + \sigma_v^2 \mathbf{I} \quad \text{and} \quad R_{sy} = \begin{bmatrix} 0 & \dots & 0 & \sigma_s^2 & 0 & \dots & 0 \end{bmatrix} H^* \quad (5.15)$$

Observe that R_{sy} is proportional to the $(\Delta + 1)$ -th row of H^* . Substituting (5.15) into (5.12) we arrive at the following expression for the equalizer tap vector:

$$w^{o*} = \sigma_s^2 e_\Delta^* H^* (\sigma_s^2 H H^* + \sigma_v^2 \mathbf{I})^{-1} \quad (5.16)$$

where e_Δ denotes the basis vector with Δ leading zeros,

$$e_\Delta \triangleq [0 \ \dots \ 0 \ 1 \ 0 \ \dots \ 0]^\top \quad (\text{with } \Delta \text{ leading zeros})$$

Moreover, according to Thm. 3.1, the resulting minimum mean-square error is given by $\text{m.m.s.e} = \sigma_s^2 - R_{sy} R_y^{-1} R_{ys}$ and, hence,

$$\text{m.m.s.e} = \sigma_s^2 (1 - \sigma_s^2 e_\Delta^* H^* R_y^{-1} H e_\Delta) \quad (5.17)$$

An intuitive way to understand the usefulness of using a nonzero delay Δ is to recall that channels have group delays. Loosely speaking, the group delay of a channel is a measure of the amount of delay that a signal undergoes when transmitted through the channel. For this reason, the channel output, $\mathbf{z}(i)$, would be more correlated with a delayed version of $\mathbf{s}(i)$, than with $\mathbf{s}(i)$ itself. It therefore makes sense to use the channel output to estimate a delayed replica of the input.

Example 5.1 (Numerical illustration)

Let us use the above results to re-examine Ex. 4.1, where $L = 3$, $M = 2$, $\Delta = 0$, $\{c(0), c(1)\} = \{1, 0.5\}$, and $\{\sigma_s^2, \sigma_v^2\} = \{1, 1\}$. Therefore, for this case, we have

$$H = \begin{bmatrix} 1 & 0.5 & & \\ & 1 & 0.5 & \\ & & 1 & 0.5 \end{bmatrix}$$

so that from (5.15),

$$R_y = \begin{bmatrix} 9/4 & 1/2 & 0 \\ 1/2 & 9/4 & 1/2 \\ 0 & 1/2 & 9/4 \end{bmatrix} \quad \text{and} \quad R_{sy} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Using (5.16) and (5.17) we get

$$w^{o*} = \begin{bmatrix} 0.4688 & -0.1096 & 0.0244 \end{bmatrix} \quad \text{and} \quad \text{m.m.s.e} = 0.5312$$

which are the same results from Ex. 4.1. ◇

5.5 APPLICATION: MULTIPLE-ANTENNA RECEIVERS

Let us reconsider Ex. 4.1 with N noisy measurements,

$$\mathbf{y}(i) = \mathbf{x} + \mathbf{v}(i), \quad i = 0, 1, \dots, N-1$$

of some zero-mean random variable $\mathbf{x}$. Let $\mathbf{y} = \text{col}\{\mathbf{y}(0), \mathbf{y}(1), \dots, \mathbf{y}(N-1)\}$ denote the observation vector. In Ex. 4.1 we evaluated the linear least-mean-squares estimator of $\mathbf{x}$ given $\mathbf{y}$ by computing $R_{xy} R_y^{-1}$ explicitly. Here we evaluate $\hat{\mathbf{x}}$ by showing first that $\mathbf{y}$

and x are related through a linear model as in (5.1), and then using (5.5). For generality, we shall assume that the variances of x and $v(i)$ are σ_x^2 and σ_v^2 , respectively. In Ex. 4.1, we used $\sigma_x^2 = \sigma_v^2 = 1$.

Introduce the $N \times 1$ column vectors:

$$\mathbf{v} \triangleq \text{col}\{v(0), v(1), \dots, v(N-1)\}, \quad \mathbf{h} \triangleq \text{col}\{1, 1, \dots, 1\}$$

Then $\mathbf{y} = \mathbf{h}x + \mathbf{v}$ and the covariance matrix of $\mathbf{v}$ is $\sigma_v^2 \mathbf{I}$. We now obtain from (5.5) that

$$\hat{x} = [1/\sigma_x^2 + \mathbf{h}^* \mathbf{h} / \sigma_v^2]^{-1} \mathbf{h}^* \mathbf{y} / \sigma_v^2 = \frac{1}{N + \frac{1}{\text{SNR}}} \sum_{i=0}^{N-1} y(i)$$

where $\text{SNR} = \sigma_x^2 / \sigma_v^2$. Observe that we are not dividing by the number of observations (which is N) but by $N + 1/\text{SNR}$. We shall comment on the significance of this observation in the next chapter.

Compared with the solution of Ex. 4.1, observe how the expression for $\hat{x}$ is obtained here more immediately. We did not explicitly form the $N \times N$ covariance matrix R_y and the $1 \times N$ cross-covariance vector R_{xy} , and then evaluate $R_{xy} R_y^{-1}$. Instead, we used the linear relation $\mathbf{y} = \mathbf{h}x + \mathbf{v}$ and the formula (5.5). In this formula, the term $(R_x^{-1} + \mathbf{H}^* R_v^{-1} \mathbf{H})$ is a scalar and its inversion is trivialized. This is in contrast to the term $(R_v + \mathbf{H} R_x \mathbf{H}^*)$, whose inverse appears in the alternative formula (5.3) — this term is equal to R_y . Again, if we interpret the $\{y(k)\}$ as noisy measurements that are collected at multiple antennas as a result of transmitting a signal x over additive noise channels, then we find that the expression for $\hat{x}$ suggests the optimal linear receiver structure shown in Fig. 5.4.

In Probs. II.25 and II.26 we extend this result further by showing how to incorporate channel gains into the design procedure. Specifically, we pursue receiver structures that are optimal according to two criteria: in one criterion, the SNR at the output of the receiver is maximized (resulting in the so-called maximal-ratio-combining technique), while in the second criterion the same minimum mean-square-error design of the current example is used.

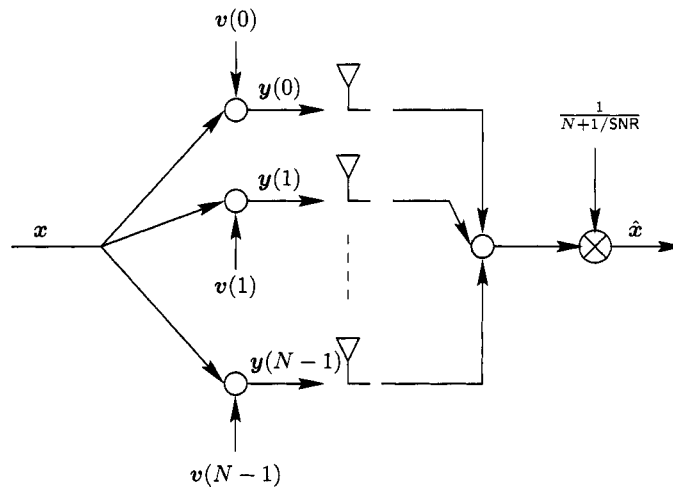


FIGURE 5.4 An optimal linear receiver for recovering a symbol x transmitted over additive-noise channels from multiple-antenna measurements.